CISS240: Introduction to Programming
Quiz q2101

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This is a closed-book, no compiler, 5 minute quiz.

The following questions involve writing function prototypes. Warning: Use `&` only when absolutely necessary.

Q1. The function `swap` swaps the values of two `int` variables. For instance if `x` is `0` and `y` is `1`, on returning from `swap(x, y)`, `x` is `1` and `y` is `0`. Write down the function prototype of `swap`. Do not include variable names in the function prototype.
Answer:

```
void swap(int&, int&);
```

Q2. The function `swap_double` swaps the values of two `double` variables. For instance is `x` is `0.0` and `y` is `1.1`, on returning from `swap_double(x, y)`, `x` is `1.1` and `y` is `0.0`. Write down the function prototype of `swap_double`. Do not include variable names in the function prototype.
Answer:

```
void swap_double(double&, double&);
```

Q3. Let `x` and `y` be integer variables. On calling `f(x, y)`, the average (as a `double` value) of `x` and `y` is returned, `x` is set to `0`, `y` is unchanged. Write down the function prototype of `f`. Do not include variable names in the function prototype.
Answer:

```
double f(int&, int);
```

Q4. If `x` is an integer array, `n` is an integer variable, `value` is an integer variable, then on calling `fill(x, n, value)`, the function will set `x[0]` to `value`, `x[1]` to `value`, `x[2]` to `value`, ..., `x[n - 1]` to `value`. Write down the function prototype of `fill`. You must include variable names in the function prototype (use `x`, `n`, `value` for their names).
Answer:

```
void fill(int x[], int n, int value);
```

Q5. The function `play_audio()` plays an audio file. The parameters are:

- `filename`: a C-string that is the name of the audio file to be loaded.
- `track_number`: an integer. 0 is the first track, 1 is the second track, etc. -1 is "play all tracks".
- `repeat`: a boolean. Track(s) will be played in a loop if `repeat` is `true`.

The function returns a 0 if there are no errors, a -1 if the file cannot be found using the `filename`, -2 if the `track_number` is not -1 and the track number is not found in the audio file. Write down the prototype of this function. You must include variable names in the function prototype (use `filename`, `track_number`, `repeat` for their names).

Answer:

```
int play_audio(const char* filename, int track_number, bool repeat);
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.